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High-speed running distance and frequency in football training: When and how are they stimulated in a microcycle?

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Abstract

Purpose: The aims of this study were to describe the distribution of the distance and frequency of the high-speed running performed by professional players during the training process, taking into account the duration of the microcycle (Mi) composed of 4, 5, 6 and 7 days between matches (MD). Total distance at high speed (HSR, >21 km·h^{-1}, in m), at very high-speed running (VHSR, >24 km·h^{-1}, in m) and sprint (SPR, >27 km·h^{-1}, in m) were recorded, in terms of distance (d) and frequency (n) during training sessions. Training data were obtained using global positioning technology from 23 professional football players belonging to the first team of the Spanish La Liga club during the 2017/18 season (n = 28 microcycles). There were no significant differences in volume (distance and frequency) of the different locomotive demands (LD) when Mi accumulated more days. Regardless of the duration of the Mi, the central days (MD-5, MD-4 and MD-3) accumulated more stimulus in HSR, VHSR and SPR compared to the post sessions (MD-6 and MD-7) and pre-match (MD-2 and MD-1) where the MD-2 was the day that required the least stimuli from the players ($p < 0.05$). The Mi7 presented significantly higher external load values in the MD-5 and MD-4 compared to the Mi5 and Mi4, where the MD-3 was the day that received the greatest stimulation in the locomotive demand ($p < 0.01$). The data demonstrate that the LD (HSR, VHSR and SPR), accumulated greater stimulus in the central days of Mi, far from the previous competition, favoring an optimal recovery without compromising performance for the next game.

Keywords

Global positioning system, locomotive demands, periodization, soccer

Introduction

Football is an invasion team sport in which player activity is unpredictable, multi-directional and intermittent.[1] Over the 90 min of a match, professional soccer players perform more than 40 physical efforts at over 21 km·h^{-1}, (almost 10% of the total distance covered - from 9 to 14 Km), interspersed with lower intensity activity.[2] Of the thousands of intermittent actions or efforts carried out during a match, between 17 and 81 are in the form of sprints (SPR [>24 Km·h^{-1}].[3] The cumulative SPR distance makes up only 3–5% of the total distance, approximately 300–500 m, amassed in repeated actions of between 10 and 15 m.[4]

Furthermore, it is worth drawing attention to two aspects relating to this type of maximum demand efforts: on the one hand, competition over the last few years has demanded ever greater distances (2%) and an increased number of efforts at high speed (by 30–50%);[5,6] at the same time there has been a rise in congested periods in competition (e.g. the number of consecutive matches played over short periods of time), such as two or three matches played in the space of eight days.[7] In this sense, there are important imbalances in the distribution of training frequency (F), intensity (I), time (T), type (T), volume (V) and progression (P) of training (known as the FITT-VP

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Distribution of the locomotive variables during a soccer full season

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principle)[8] within microcycle (Mi) due to the differences that exist between the player's role, position, and level of participation in competition (starters vs. non-starters). These differences could lead to important differences in soccer-specific fitness dimensions, with the differences widening as competition progresses.[9,10] Within this highly demanding conditional context, a need has arisen to develop player capacity for producing and accumulating high-speed running distances with the double aim of optimising physical performance and reducing the probability of sustaining an injury.[11] In order to respond to the demand of being able to accumulate a large quantity of distance at high speed, the proposal is for an elevated chronic FITT-VP which could have a protective effect, so that more robust players would be able to cope better with exposure to the high speed accumulated in distance and frequency.[11,12] Adequate planning and intervention in the training process, and therefore in the management of this type of stimulation is one of the pillars with which to optimise this process.

The Mi with six days between matches is the basic planning unit in performance football. However, the need to participate in various competitions at the same time, together with other non-sports related determinants (e.g. timetables established by television) may alter the duration of the Mi (e.g. from 3 to 8 days between matches). Various previous studies have described weekly patterns of four,[13–17] five[18,19] and six[7,20] training sessions between matches. Predictably, these studies agree that the higher the Mi, the greater the FITT-VP demanded of players during the week, even though no absolute values are given.[21] Furthermore, as regards the variables of high-speed movement, football teams could accumulate a high-speed running distance (HSR, >19.8 or >21.0 $km \cdot h^{-1}$) of around half a kilometre (146–811 m)[15,17] and of 8–217 m at sprint (SPR, >24.0 or 25.2 $km \cdot h^{-1}$) within a Mi of four sessions and just one match.[13,18,22,23] However, the range of speeds adopted in previous studies does not allow us to draw a clear comparison.

More recently, there has been an attempt to detail the distribution of FITT-VP carried out by professional teams in competitive Mi sessions, using Malone et al.'s.[15] suggestion of 'day post' (MD +) or 'day pre' (MD-) match. The studies appear to coincide in that professional football teams have proposed a horizontal alternation in the Mi in order to stimulate different conditional qualities and reduce physiological interference,[23,24] in an attempt to avoid over-training[25] and prevent injury.[26] The relevant literature agrees that weekly planning should be split into three phases[14,18,21,23,27]: 1) recovery sessions for those players who took most part in the match, but compensatory for the rest in which high-speed locomotive demands (LD) were unstimulated due to absence of competition;[22] 2) acquisition sessions aimed at proposing strength, resistance and speed 'training' activities.[24] For example, previous studies show that running spurts at HSR (40–390 m), VHSR (114–132 m) and SPR (6–107 m) were more stimulated on days MD-4[13,18,23] and MD-3[16,17,22,28] during the Mi. The justification for the speed stimulation in the middle days (especially MD-3) may be linked to better physical performance in the next match.[29] Phase 3 would consist of *tapering* sessions in the days prior to the match in an attempt to guarantee an adequate fine tuning ready for competition.[13,30] It would be interesting to know how teams propose the distribution of LD, especially those relating to high-speed running in the Mi, depending on the available training days between matches.

Therefore, monitoring the variations and changes of LD during the competitive week could provide sport scientists and coaches with important information about the possible differences of variation and fluctuation of the LD that exist between the different durations of Mi depending on the moment of the competitive season. Based on previous studies,[21] significant variations in weekly LD can be found depending on the moment of the season and the length of the Mi. Given the above, the aims of this study were to describe and compare how the volume and frequency of the three high-speed running variables (HSR, VHSR and SPR) (1) are accumulated in a Mi in relation to its duration (from 4 to 7 days between matches), and (2) their distribution depending on the distance between the session and the next match within the Mi (from MD-7 to MD-1).

Methods

Design

Using a descriptive research design, a team of twenty-three professional male soccer players was monitored daily throughout a one full-competitive season. Twenty-eight weeks of all training session demands (n = 140) were collected over one all competition seasons (2018–2019) and monitored using 10 Hz GPS units. For the purposes of the present study, only weeks with one official match and with four or more training sessions were included in this study. This refers to training sessions (does not include the match) in which both the starting and non-starting players trained together and complete all Mi training. This means that sessions after the match day were included whenever both starters and non-starters trained together including days without training. This decision was made to reduce the variability among comparisons. Thus, only 28 weeks' worth of data were included in the analysis. The types of Mi (used as independent variables) were classified based on the number of sessions between matches (4 days, 5 days, 6 days, and 7 days). A graphical depiction of the weekly distribution of training sessions and matches across the season is presented in Figure 1.

Participants

Twenty-three professional soccer players participated in this study (age [mean ± standard deviation] = 27 ± 3.5 years, body mass = 77.2 ± 6.6 kg, height = 181.8 ± 5.5 cm, and

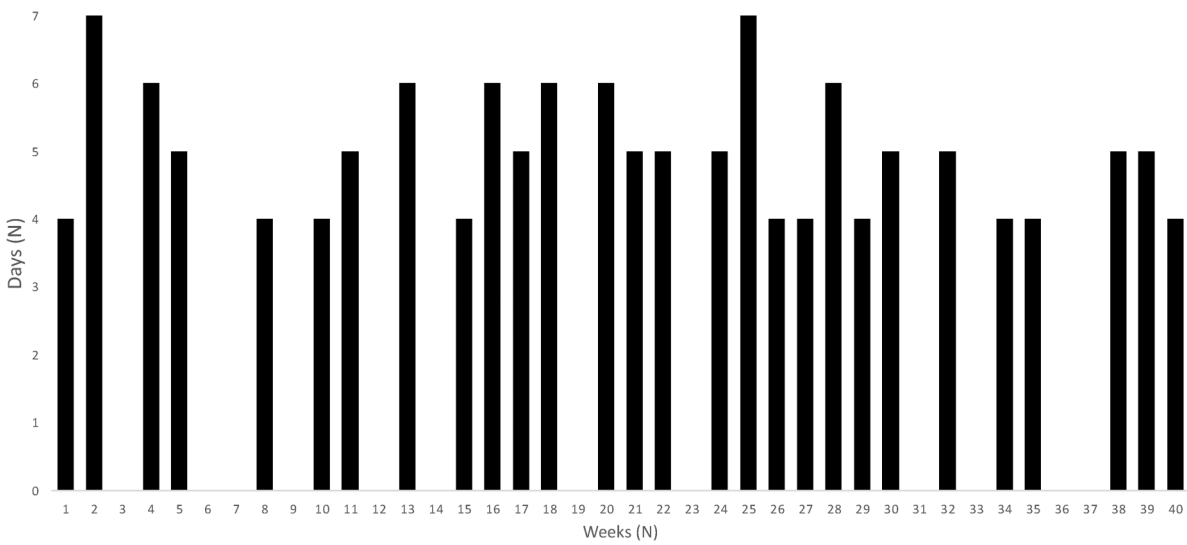

Figure 1. Timeline of the observational study and number of sessions and weeks.

fat % = 7.5 ± 1.9%). All players belonged to the same team of the Spanish first division *La Liga*. Data from 28 Mi of the competitive period during the 2018–2019 season were collected for analysis. The procedures followed in this study were in accordance with the Declaration of Helsinki and were approved by the Ethics Committee of the University of the Basque Country (UPV/EHU).

Type of microcycles

The competitive season was divided into different types of Mi for both starters and non-starters due to the irregular match schedule, which led to various weekly patterns with different numbers of (full) days between matches (number of weeks, 4 days n = 10, 5 days n = 10, 6 days n = 6, and 7 days n = 2). This division allowed us to investigate the influence of the different LD patterns incorporated within this training unit. Based on the recommendations of previous studies,[13,15,18,31] the LD of the selected Mi were categorized using the "match day minus" format (MD-). In a week with only one match, they often included 1 day off and 4–6 training sessions (day until next match: MD-6 MD-5, MD-4, MD-3, MD-2 and MD-1). The training sessions, which are contextualized below, included integrated content (e.g. tactical, technical and physical factors were combined). The regular training sessions were later classified as MD-4, MD-3, MD-2 and MD-1.[15,17] Data from preseason and training weeks during the FIFA International Match Calendar, as well as breaks and weeks with two competitive matches were excluded from the analysis.[17]

External variables

The high-speed running (HSR, >21 km·h^{-1}), very high-speed running (VHSR, >24 km·h^{-1}) and the sprint distances (SPR, >27 km·h^{-1}) were quantified in terms of distance (d in m) and frequency (n in occurrences) to the training. The speed thresholds used for determining HSR, VHSR, and SPR have been established in previous studies[21,28,31–34] LD data for each player were organized chronologically, including days without training ('0' values were included).

Procedures

Player's activity during each training session was monitored using a por table 10 Hz GPS unit (S5, Catapult Innovations Victoria, Australia). Each unit was placed inside a mini pocket positioned between the shoulder blades of a specially designed vest To avoid interunit errors, each player used the same device during the entire study.[18] After completing each training session, GPS data were extracted using the corresponding software (Openfield 2.4). The number of satellites used to infer GPS signal quality, horizontal dilution of precision and the average of the GNSS quality were: 10.1 ± 0.8 satellites, 0.9 ± 0.2 and 69.6 ± 9.3%, respectively.

Statistical analyses

Before any statistical analysis, the normal distribution of the data (Shapiro–Wilk test) and the homogeneity of variances (Levene's test) were confirmed ($P > 0.05$). Descriptive data are presented as means and standard deviations (SD) unless otherwise noted. In order to determine the impact of various periods of the Mi on the different dependent variables, we performed one-way analyses of variance with the Mi as the only within-participants factor. When applicable, Bonferroni and Dunnett's T3 post hoc tests were used to compare the differences between two periods of the Mi.

Effect sizes (ES) were calculated to determine significant differences. The magnitude of differences were reported by means of the Cohen's d (effect size = ES) with the corresponding 95% confidence intervals, and they were classified as trivial (<0.2), small (0.2–0.6), moderate (0.6–1.2), large (1.2–2.0), and very large (>2.0).[35] We also calculated the coefficient of variation (CV) in order to assess variations in the Mi.[36] To examine the differences between days of training sessions and regular games, a 4-level mixed linear model was used. MD minus was integrated into the mixed linear model as an independent categorical variable to estimate the difference in results between the different days before the match (MD-4, MD-3, MD-2 and MD-1). All statistical analyses were performed using SPSS software version 25.0 (SPSS Inc., Chicago, IL, USA) and statistical significance was set at an alpha level of 0.05.

Results

Table 1 depicts the total LD accumulated in the three variables studied (HSR, VHSR and SPR), in both distance and frequency, for each type of Mi. Results indicated that there were no significant differences in the total LD accumulate between HSR, VHSR or SPR and Mi during training.

Figures 2–4 depicts the distribution of LD values obtained from each training sessions (MD-) during each Mi. In regard to the differences between sessions in the same Mi, our data revealed that the MD-5 presented significant differences and moderate to very large effects with MD-2 in dHSR (ES = 1, 0.57/1.40) and nHSR (ES = 0.8, 0.40/1.21) within Mi7. Furthermore, the MD-5 exhibited statistically significant differences and moderate to extremely large effects with MD-2 and MD-1 within Mi6 in almost all locomotive variables dHSR (ES [range] = 0.70–1.61), nHSR (ES [range] = 1–1.62), dVHSR (ES [range] = 0.44–1.22), nVHSR (ES [range] = 0.44–1.29), dSPR (ES [range] = 0.36/0.97) and nSPR (ES [range] = 0.43–1.06).

With reference to the MD-4 presented significant differences and moderate to extremely large effects with MD-2 within Mi7, 6 and 5 in dHSR (ES [range] = 1.10–4.12), nHSR (ES [range] = 1.22/3.77), dVHSR (ES = 0.94/3.79), nVHSR (ES [range] = 0.90/3.53) and nSPR (ES [range] = 0.56/1.40). However, MD-4 presented significant differences and moderate to extremely large effect only in dSPR (ES [range] = 0.87/1.74) with MD-2 within Mi7 and Mi6. The MD-4 presented significant differences and moderate to extremely large effects with MD-7, MD-5, MD-3 in dHSR (ES = −4.7, −5.76/−3.57 & ES = −2.5, −3/1.97 & ES = 0.8, 0.36/1.20), nHSR (ES = −5, −6.15/−3.90 & ES = −2.4, −2.94/−1.93 & ES = 1.2, 0.80/1.66), dVHSR (ES = −3.7, −4.65/−2.76 & ES = −2.7, −3.25/−2.17 & ES = 1.2, 0.77/1.61), nVHSR (ES = −3.8, −4.71/−2.80 & ES = −2.8, −3.33/−2.25 & ES = 1.7, 1.25/2.15), dSPR (ES = −1.4, −2.07/−0.63 & ES = −1.2, −1.59/−0.76 & ES = 0.5, 0.11/0.89) within Mi7. Within Mi6 the MD-4 presented significant differences and moderate to extremely large effect with MD-5 and MD-1 in dHSR (ES [range] = 1.10/2.19), dVHSR (ES = −1.3, −1.53/−1.01 & ES = 0.9, 0.66/

Table 1. Descriptive statistics of locomotive variables of regular training (mean ± SD).

Variables	Mi7	Mi6	MI5	Mi4	*p*	d
dHSR (m)	546.6 ± 100.0	504.4 ± 140.5	397.7 ± 141.1	358.6 ± 97.7	Mi4 vs Mi7: 0.001	Mi4 vs Mi7: 0.353
					Mi5 vs Mi7: 0.014	Mi5 vs Mi7: 0.247
					Mi6 vs Mi7: 0.018	Mi6 vs Mi7: 0.238
nHSR (n)	40.06 ± 4.74	36 ± 9.4	30.03 ± 10.3	28.1 ± 7.8	Mi4 vs Mi7: 0.007	Mi4 vs Mi7: 0.363
					Mi5 vs Mi7: 0.037	Mi5 vs Mi7: 0.305
					Mi6 vs Mi7: 0.016	Mi6 vs Mi7: 0.312
dVHSR (m)	208.6 ± 54.4	161.6 ± 65.8	140.8 ± 67.5	105.3 ± 37.5	Mi4 vs Mi5: 0.004	Mi4 vs Mi7: 0.209
					Mi4 vs Mi7: 0.000	Mi5 vs Mi7: 0.485
					Mi5 vs Mi7: 0.003	Mi5 vs Mi7: 0.268
					Mi6 vs Mi7: 0.000	Mi6 vs Mi7: 0.293
nVHSR (n)	13.65 ± 2.7	10.1 ± 3.4	9.8 ± 4.5	7.02 ± 2.6	Mi4 vs Mi7: 0.001	Mi4 vs Mi7: 0.224
					Mi4 vs Mi7: 0.000	Mi4 vs Mi7: 0.492
					Mi6 vs Mi7: 0.003	Mi6 vs Mi7: 0.338
dSPR (m)	40.7 ± 22.0	40.7 ± 27.0	30.5 ± 18.0	17.4 ± 8.0	Mi4 vs Mi6: 0.007	Mi4 vs Mi7: 0.227
					Mi4 vs Mi7: 0.000	Mi4 vs Mi7: 0.315
nSPR (n)	2.22 ± 1.3	1.94 ± 1.0	1.85 ± 1.0	0.86 ± 0.3	Mi4 vs Mi5: 0.012	Mi4 vs Mi5: 0.245
					Mi4 vs Mi6: 0.046	Mi4 vs Mi6: 0.220
					Mi4 vs Mi7: 0.001	Mi4 vs Mi7: 0.295

Note: Mi7: microcycle 7; Mi6: microcycle 6; Mi5 : microcycle 5; Mi4 : microcycle 4; dHSR = total distance above 21 km·h^{-1}; nHSR = number of high speeds running above 21 km·h^{-1}; dVHSR = total distance above 24 km·h^{-1}; nVHSR = number of high speeds running above 24 km·h^{-1}; dSPR = total distance above 27 km·h^{-1}; nSPR = number of high speeds running above 27 km·h^{-1}.

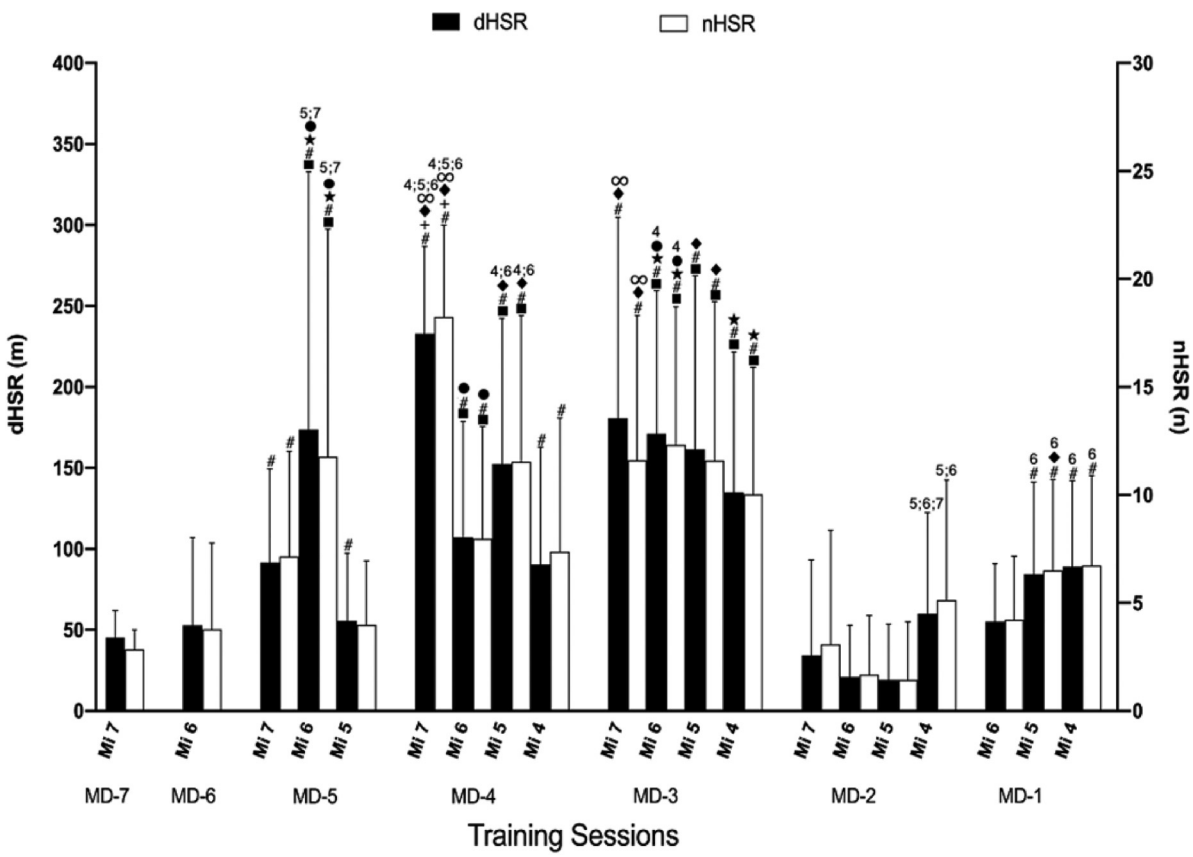

Figure 2. Locomotive variables (mean and standard deviation) for professional players relative to competitive match play. Data are presented across in volume (meters: m) and frequency (number: n): dHSR = total distance above 21 km·h^{-1}; nHSR = number of high speeds running above 21 km·h^{-1}. Mi7 is microcycle 7, Mi6 is microcycle 6, Mi5 is microcycle 5 and Mi4 is microcycle 4. For the comparison of MD- within the same Mi ($p < 0.05$): ∞ is >MD-7, ● is >MD-6, ◆ is >MD-5, * is >MD-4, + is >MD-3, # is >MD-2, and ■ is >MD-1. For the comparison of the same MD- between Mi: 1 is >Mi1, 2 is >Mi2, 3 is >Mi3, 4 is >Mi4, is, 5 is >Mi5, 6 is >Mi6 and 7 is >Mi7.

1.10), nVHSR (ES = −1.2, −1.49/−0.96 & ES = 0.8, 0.59/1.02) and dSPR presented difference only with MD-1 (ES = 0.7, 0.44/0.86). Finally, within Mi4 the MD-4 exhibited statistically significant differences and moderate to large effect with MD-2 in dVHSR (ES = 0.7, 0.51/0.91), nVHSR (ES = 0.6, 0.40/0.80).

In relation to the MD-3 presented significant differences and moderate to extremely large effects with MD-7 and MD-2 in dHSR (ES = −1.6, −2.33/−0.85 & ES = 1.5, 1.08/1.99), nHSR (ES = −1.8, −2.58/−1.09 & ES = 1.4, 0.98/1.85), dVHSR (ES = −1.5, −2.26/−0.81 & ES = 1.3, 0.85/1.71), nVHSR (ES = −1.8, −2.56/−1.07 & ES = 1.14, 0.72/1.56), and only for MD-2 in dSPR (ES = 0.7, 0.33/1.15) within Mi7. Furthermore, the MD-3 presented significant differences and moderate to extremely large effects with MD-2 and MD-1 within Mi6, 5 and 4 in almost all locomotive variables dHSR (ES [range] = 0.46/2.55), nHSR (ES [range] = 0.46/2.47), dVHSR (ES [range] = 0.7/2.02), nVHSR (ES [range] = 0.47/1.90), dSPR (ES [range] = 0.40/1.42), nSPR (ES [range] = 0.34/1.30). Additionally, the MD-3 presented significant differences and moderate to extremely large effects with MD-6 within Mi6 and with MD-5 within Mi5 in almost all locomotive variables dHSR (ES [range] = −1.54/−1.33), nHSR (ES [range] = −1.60/−1.35), dVHSR (ES [range] = −1.49/−1.25), nVHSR (ES [range] = −1.52/−1.08), dSPR (ES [range] = −1.17/−0.79), nSPR (ES = 0.9, 0.64/1.11 & ES = −0.5, −0.77/−0.30). Finally, the MD-1 presented significant differences and very large to extremely large effect with MD-2 in nHSR (ES = 1.4, 1.20/1.66).

In relation to the differences in the same MD- session between the different Mis, the results were that MD-5 presented significant differences and moderate to extremely large effect between Mi6 versus Mi5 in almost all locomotive variables dHSR (ES = 1, 0.68/1.28), nHSR (ES = 1, 0.71/1.30), dVHSR (ES = 0.8, 0.49/1.05), nVHSR (ES = 0.8, 0.54/1.12), dSPR (ES = 0.6, 0.33/0.91) and nSPR (ES = 0.3, 0.06/0.62). The MD-4 exhibited significant differences and moderate to extremely large effect between

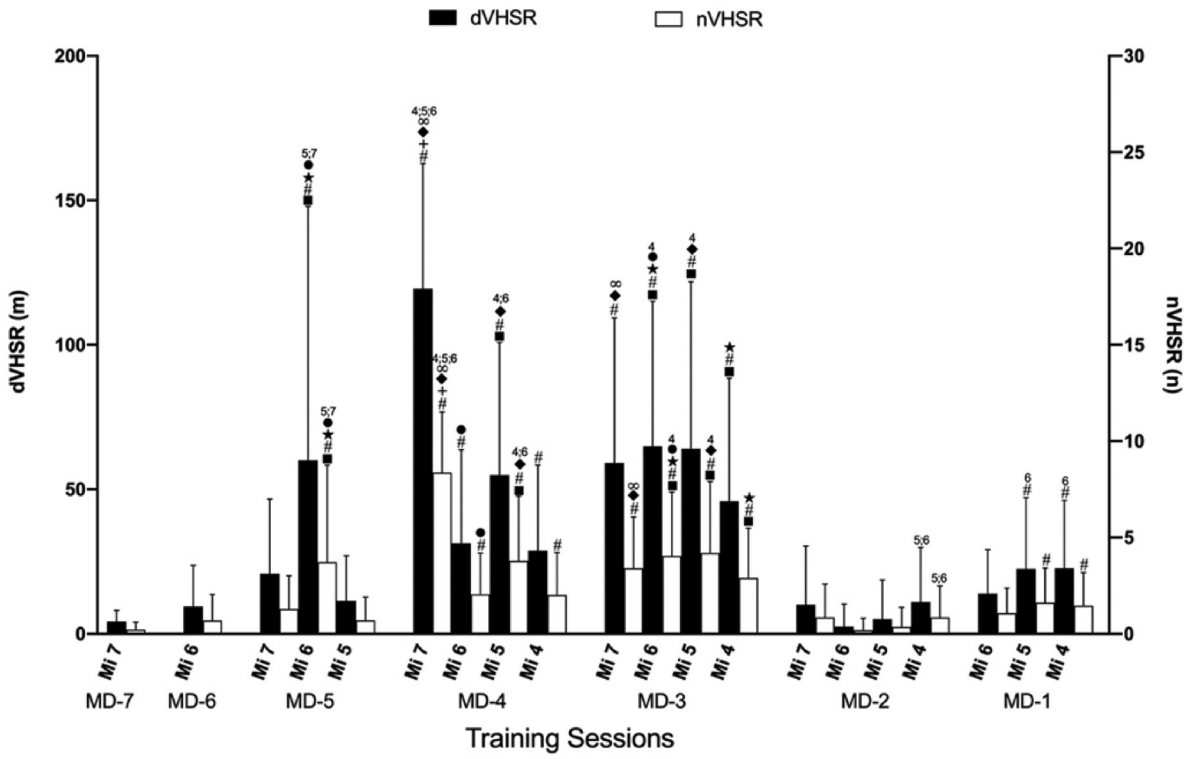

Figure 3. Locomotive variables (mean and standard deviation) for professional players relative to competitive match play. Data are presented across in volume (meters: m) and frequency (number: n): dVHSR = total distance above 24 km·h^{-1}; nVHSR = number of high speeds running above 24 km·h^{-1}. Mi7 is microcycle 7, Mi6 is microcycle 6, Mi5 is microcycle 5 and Mi4 is microcycle 4. For the comparison of MD- within the same Mi ($p < 0.05$): ∞ is >MD-7, ● is >MD-6, ♦ is >MD-5, * is >MD-4, + is >MD-3, # is >MD-2, and ■ is >MD-1. For the comparison of the same MD- between Mi: 1 is >Mi1, 2 is >Mi2, 3 is >Mi3, 4 is >Mi4, is, 5 is >Mi5, 6 is >Mi6 and 7 is >Mi7.

Mi7 versus Mi6, Mi5 and Mi4 in almost all locomotive variables dHSR (ES [range] = 0.72/2.57), nDHSR (ES [range] = 0.86/2.56), dVHSR (ES [range] = 1.05/2.78), nVHSR (ES [range] = 1.07/2.75), dSPR (ES [range] = 0.39/1.38), nSPR (ES = 0.7, 0.18/2.78). Additionally, the MD-4 presented significant differences and moderate to large effect between Mi5 versus Mi4 in dHSR (ES = 0.8, 0.54/0.97), nHSR (ES = 0.6, 0.43/0.85), dVHSR (ES = 0.7, − 0.47/0.89), nVHSR (ES = 0.6, 0.41/0.84) and between Mi5 versus Mi6 in dVHSR (ES = − 1.4, − 1.69/ −1.19).

Discussion

The aim of this study was to describe and compare the weekly distribution of the distance and frequency of three high-speed running variables (HSR, VHSR and SPR) depending on the duration of the Mi (4, 5, 6 and 7 days between matches) and on the distance (in days) between the session and the match within the Mi. The most relevant results obtained from the study were that a greater LD was not accumulated in the three running variables (HSR, VHSR and SPR) when the Mi was longer and, the demand for these variables within the Mi was concentrated in the middle days (e.g. MD-3 and MD-4). However, not all the types of Mi analysed had a similar distribution in the same days in relation to the following match. Finally, it is remarkable that there was substantial variability in training outputs regarding sprint quantity among players. The value of the standard deviation in each training was high.

Unlike previous research works which detected that the longer the duration of the Mi the greater the FITT-VP,[21,37] in this study we did not find significant differences in the locomotive variables studied (Table 1). This lack of agreement in the results could be explained by a number of things, one of which may be the difference in the speed thresholds used in the aforementioned research works (HSR: >14.4 km·h^{-1}, in Curtis et al.,[37] and HSR: 20.0–24.9 km·h^{-1} in Clemente et al.[21] and SPR: >25.0km·h^{-1} in Clemente et al.[21] compared with those used in this study. However, we believe one explanation for the differences could be the particular planning strategy used by staff in the distribution of the running variables during the Mi depending on the available days between matches.[38]

Regardless of the duration of the Mis, it was on the mid-week days that the biggest FITT-VP (distance and

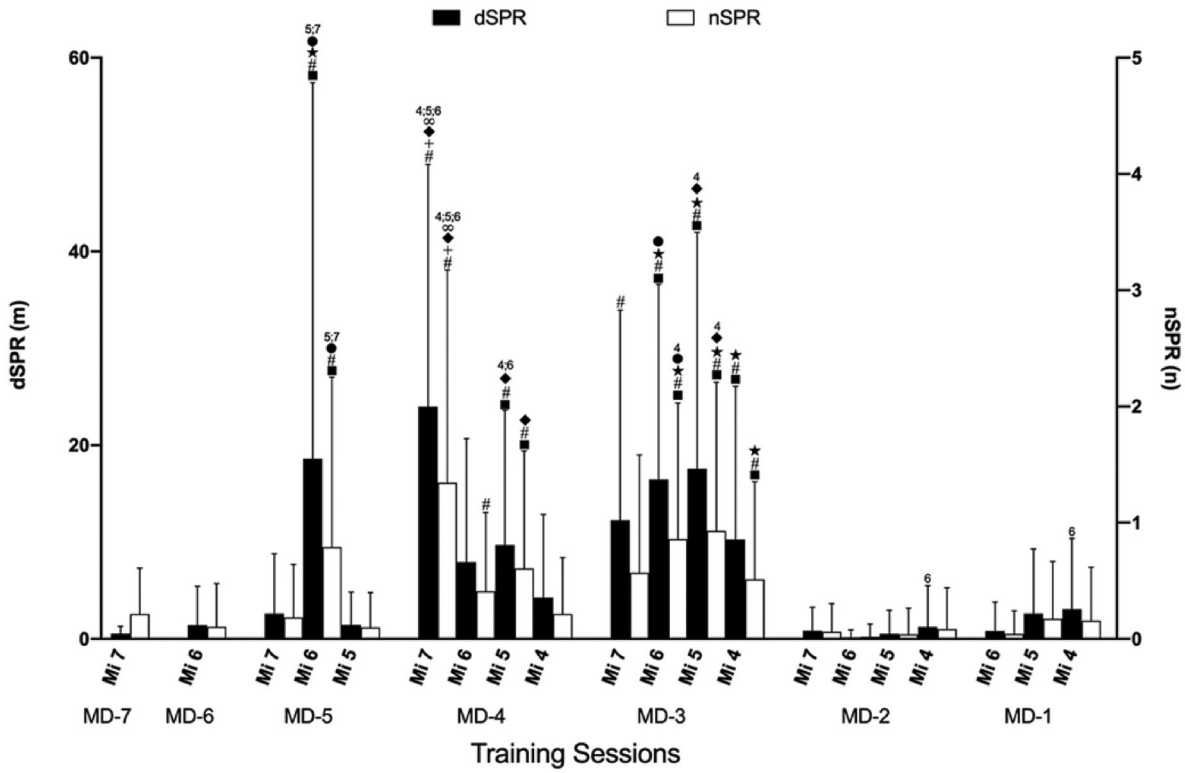

Figure 4. Locomotive variables (mean and standard deviation) for professional players relative to competitive match play. Data are presented across in volume (meters: m) and frequency (number: n): dSPR = total distance above 27 km·h^{-1}; nSPR = number of high speeds running above 27 km·h^{-1}. Mi7 is microcycle 7, Mi6 is microcycle 6, Mi5 is microcycle 5 and Mi4 is microcycle 4. For the comparison of MD- within the same Mi ($p < 0.05$): ∞ is >MD-7, ● is >MD-6, ♦ is >MD-5, * is >MD-4, + is >MD-3, # is >MD-2, and ■ is >MD-1. For the comparison of the same MD- between Mi: 1 is >Mi1, 2 is >Mi2, 3 is >Mi3, 4 is >Mi4, is, 5 is >Mi5, 6 is >Mi6 and 7 is >Mi7.

frequency) was accumulated in the high-speed variables in relation to the days immediately after the previous match (MD-7 and MD-6) and before the match of the week in progress (MD-2 and MD-1), which is similar to what has previously been described.[13,15,18,39] This planning with an inverted U profile coincides with the literature in which the use of a horizontal alternation of the weekly training programme has been justified.[24] This planning of the Mi gives rise to a high stimulation of this type of variable via acquisition sessions (on the mid-week days) with the aim of developing and/or maintaining different physical capacities (strength, resistance and speed) throughout the season.[40] Furthermore, variation in the content of the FITT-VP (e.g. cardiovascular, neuromuscular or locomotive demand) together with quantity (e.g. overtraining or recuperation sessions) is an interesting match approach strategy with the aim of achieving maximum performance and attempting to prevent the phenomenon of interference between different capacities.[40,41] Stimulating different conditional aspects during the mid-week acquisition sessions of competition week would avoid the overstimulation of certain variables that may compromise availability for competition at the end of the Mi.[24]

Previous studies have described a decrease in FITT-VP especially on MD-1[15,22,25] suggesting the need for a *tapering* phase prior to the next match.[13] In this research, it was the MD-2 session that demanded least from the players, regardless of the duration of the Mi. This difference can be justified by the peculiarity of the planning proposal used by the team, which is different from other proposals such as the periodization tactic[24] or in the structured Mi[18] where MD-2 is not the session with the least training stimulus.

Regarding the comparative of the same training days (MD-) in Mis of different durations, it should be pointed out that there were significant differences in the stimulation of the HSR, VHSR and SPR variables. In line with previous studies,[21] we observed that in Mi7 (long microcycle) the values of the locomotive variables studied were statistically higher on days MD-5 and MD-4 compared with the shorter duration Mis (e.g. Mi5 and Mi4). Therefore, when the team had a higher number of sessions between matches at its disposal it planned these sessions with a greater high-speed LD, distancing them as far as possible from the following match with the aim of optimising the biological adaptations effective for maximum performance.[41,42] That is, with a

higher number of available sessions between matches the team stimulated high speed in the mid-week sessions, but further away from the next match (e.g. MD-5 and MD-4), unlike the shorter sessions where MD-3 was chosen to stimulate this locomotive dimension. As suggested by Grünbichler et al.[29] it would appear that over-stimulation of high-speed locomotive demand 72 h prior to competition could improve workload efficiency (e.g. estimated distance divided by an internal load indicator or TRIMP), in accordance with the over-compensation principle.

Regarding players sprint demands, it cannot be avoided to mention that the intra-session variability was substantial. These deviations could be justified for different reasons. Someone could be due to fitness status,[43] position[31,43,44] of the players or moment of the season,[43] that they have a particular stimulation according to the match requirements. Also, more or less participation in competition during the season could have influence in the adequacy of some task during the week[44] in order to facilitate the recovery or preserve for the next match.

While this research might be original in its description of the distribution of locomotive variables HSR, VHSR and SPR in different types of competitive football Mis for the same elite team, it is not without its limitations. Firstly, a generalisation of the results should be treated with caution, due to these depending, above all, on the training philosophy established by staff.[18,20,32,45] Lastly, it would have been interesting to include some kind of locomotive metrics (e.g. objective derived from heart rate or subjective as a response to the perception of the effort), with which to discover the players' response and assess their adaptation process and availability for competition. Finally, the inclusion of information about some aspects such us testing, readiness for competition or physical performance in competition, among others, could be give us more evidence about if the quantity and the distribution of the sprint demands are enough to optimize the fitness of the players and at the same time to promote mithridatism.

Practical applications

Our results underline the importance of stimulating high-speed running (e.g. HSR, VHSR and SPR) in the training process, especially on mid-week days, thus distancing it from both the previous match (allowing the players to recover) and the following one (not jeopardizing their competition performance) regardless of the duration of the Mi. It would seem appropriate for the technical team to consider dealing with the planning of these kinds of training stimulus, contextualising them to the number of available sessions between matches in an attempt to optimize player performance. This would avoid, on the one hand insufficient stimuli that would not produce the desired adaptations which could lead to a lack or loss of training and, on the other hand, an excessive or over-stimulation that would provoke fatigue or over-training, increasing the probability of injury and/or the player not being fresh for the next match.[30]

Conclusion

This research describes the strategy used by a professional football team in the planning of competitive Mis and arrives at a double conclusion. Firstly, there were no significant differences between the accumulation of distance and frequency of the high range speed variables (>21 km·h^{-1}) regardless of the number of sessions between matches - representative of a vertical homogeneity - or between successive weeks (inter-Mi) of the competition period. Secondly, as an intra-Mi strategy, a horizontal alternation was chosen; that means in the training process proposing a greater stimulation of LD in high-speed ranges (HSR, VHSR and SPR) on the mid-week days, distancing them as far as possible from the next match.

The manuscript has been read and approved by all the listed co-authors and meets the requirements of co-authorship as specified in the "Authorship Guidelines for International Journal of Sport Science & Coaching"

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